

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What is a standard atmosphere? What are the standard values at sea level? Explain how does an airplane fly.
- Q.24 Describe the working of flaps, slats and trim tab control.
- Q.25 Describe the working of hydraulic system with the function of each component explained.

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1st Sem / Aeronautical Engg.

Subject : Basics of Aeronautics

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The lift to drag ratio is maximum in the case of _____
- a) Fighter airplane b) DRONE
c) Transport aircraft d) Glider
- Q.2 The cross-sectional shape of wing is called _____
- a) Airfoil b) Circle
c) Camber d) Chord
- Q.3 The critical Mach number is _____ drag divergence number
- a) Less than b) More than
c) Equal to d) Not relevant to

- Q.4 In climbing flight, the lift is _____ than weight?
- a) Less b) Equal to
c) More d) Can not say
- Q.5 If the aircraft has a power-producing engine which have driven propellers then the power is to be converted to thrust to formulate the equations of motion.
- a) True b) False
- Q.6 What is a camber line
- a) Distance between leading edge and chord
b) Distance between chord and chamber
c) Midpoint of the upper and lower surface points
d) Distance between trailing edge and chord

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 What is the temperature of outside an airliner flying at 5000m altitude?
- Q.8 What is the use of oxygen system in an aircraft?
- Q.9 Which type of engine can fly in space?

- Q.10 What is a longeron?
- Q.11 What is a critical Mach number?
- Q.12 Turbojet is more/less quite as compared to turbofan engine.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is the difference between IC engine and turbo engine?
- Q.14 What are the structural components of a wing?
- Q.15 What is the function of trim tab in the airplane?
- Q.16 What is coefficient of lift for airfoil and wing?
- Q.17 Describe the various forces acting on an airplane.
- Q.18 Explain any one hydraulic system of aircraft.
- Q.19 What are various types of controls for surfaces?
- Q.20 What are various primary and secondary control surfaces in an airplane?
- Q.21 How the helicopter motion is controlled?
- Q.22 What is the function of skin of the aircraft?